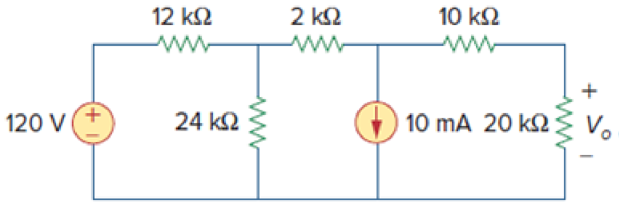
Chapter 4, Problem 56P

Problem

Use Norton's theorem to find *Vo* in the circuit of Fig. 4.122.

Figure. 4.122:



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 4.122 in the text book.

Disconnect the load resistance and short the output terminals to calculate the Norton's current.

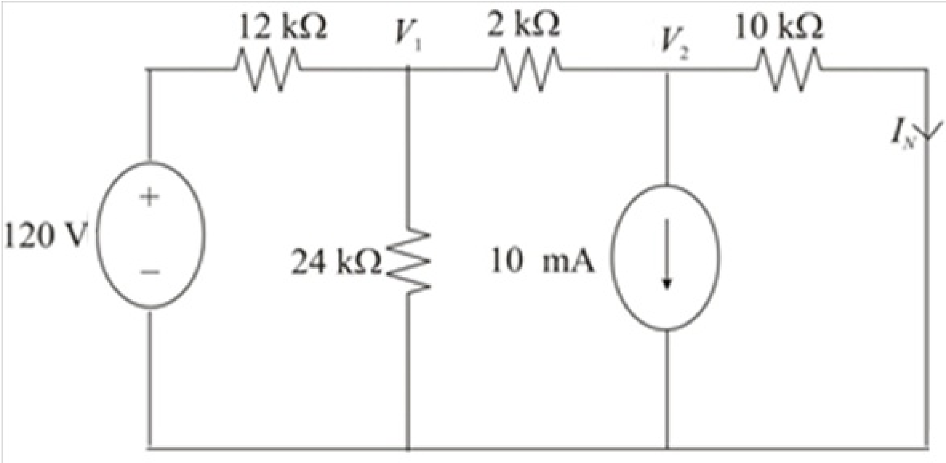


Figure 1

Step 2 of 8

Apply Kirchhoff's current law at node 1.

$$\frac{V_1-120}{12\text{ k}}+\frac{V_1}{24\text{ k}}+\frac{V_1-V_2}{2\text{ k}}=0$$

$$\frac{2V_1-240+V_1+12V_1-12V_2}{24\text{ k}}=0$$

$$15V_1-240-12V_2=0$$

$$15V_1=240+12V_2$$

$$V_1=\frac{240+12V_2}{15}$$

$$V_1=\frac{80+4V_2}{5} \quad \text{...... (1)}$$

Step 3 of 8

Apply Kirchhoff's current law at node 2.

$$10 \text{ m} + \frac{V_2 - V_1}{2 \text{ k}} + \frac{V_2}{10 \text{ k}} = 0$$

$$\frac{100 + 5V_2 - 5V_1 + V_2}{10 \text{ k}} = 0$$

$$100 + 6V_2 - 5V_1 = 0$$

Substitute $\frac{80 + 4V_2}{5}$ for V_1.

$$100 + 6V_2 - 5\left(\frac{80 + 4V_2}{5}\right) = 0$$

$$100 + 6V_2 - 80 - 4V_2 = 0$$

$$2V_2 + 20 = 0$$

$$2V_2 = -20$$

$$V_2 = -\frac{20}{2}$$
$$= -10 \text{ V}$$

Step 4 of 8

Calculate the value of the Norton's current, I_N.

$$I_N = \frac{V_2}{10 \text{ k}}$$

Substitute -10 V for V_2.

$$I_N = \frac{-10 \text{ V}}{10 \text{ k}\Omega}$$
$$= -1 \text{ mA}$$

Therefore, the value of the Norton's current I_N is $\boxed{-1 \text{ mA}}$.

Step 5 of 8

Set the value of the sources to zero to calculate the Norton's resistance.

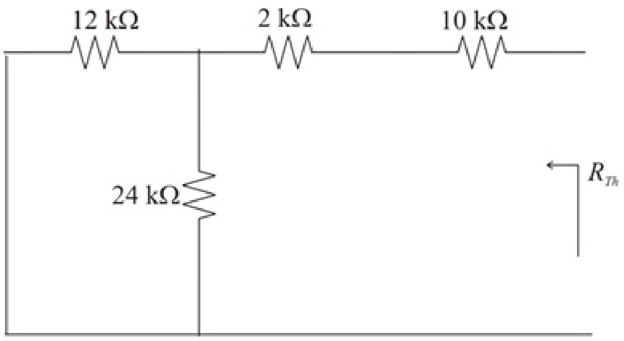


Figure 2

Step 6 of 8

Calculate the value of the Norton's resistance R_N.

$$\begin{aligned} R_N &= 10\ \text{k}\Omega + 2\,\text{k}\Omega + (12\ \text{k}\Omega)\,||\,(24\ \text{k}\Omega) \\ &= 12\ \text{k}\Omega + \frac{(12\ \text{k}\Omega)(24\ \text{k}\Omega)}{12\ \text{k}\Omega + 24\ \text{k}\Omega} \\ &= 12\ \text{k}\Omega + 8\ \text{k}\Omega \\ &= 20\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the Norton's resistance R_N is $\boxed{20\ \text{k}\Omega}$.

Step 7 of 8

Draw the Norton's equivalent circuit diagram.

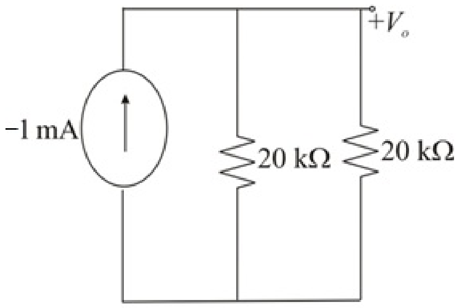


Figure 3

Step 8 of 8

Calculate the value of the voltage, V_o using current division rule.

$$\begin{aligned} V_o &= (-1\ \text{mA})(20\ \text{k}\Omega)\,||\,(20\ \text{k}\Omega) \\ &= (-1\ \text{mA})\left(\frac{(20\ \text{k}\Omega)(20\ \text{k}\Omega)}{20\ \text{k}\Omega + 20\ \text{k}\Omega}\right) \\ &= (-1\ \text{mA})\left(\frac{400\ \text{k}\Omega}{40}\right) \\ &= (-1\ \text{mA})(10\ \text{k}\Omega) \end{aligned}$$

$$= -10\ \text{V}$$

Therefore, the value of the voltage V_o is $\boxed{-10\ \text{V}}$.